

RAK12012 WisBlock Heart Rate Sensor Module Datasheet

Overview

Description

The RAK12012, a part of WisBlock Sensor, is an integrated pulse oximetry and heart-rate monitor module used for measuring person's heart rate and oxygen saturation. The sensor attached to this module is MAX30102 from Analog Devices.

Features

- Heart-Rate Monitor and Pulse Oximeter Sensor in LED Reflective Solution
- Ultra-Low Power Operation
- Robust Motion Artifact Resilience
- Fast Data Output Capability
- -40 °C to +85 °C Operating Temperature Range
- 3.3 V Power supply
- Operating Current: 0.7 - 1200 uA
- Chipset: Analog Devices MAX30102
- Module Size: 25 mm x 35 mm

Specifications

Overview

Mounting

The RAK12012 WisBlock Heart Rate Sensor Module can be mounted to the IO slot of the [WisBlock Base](#)  board. **Figure 1** shows the mounting mechanism of the RAK12012 on a WisBlock

Base module.

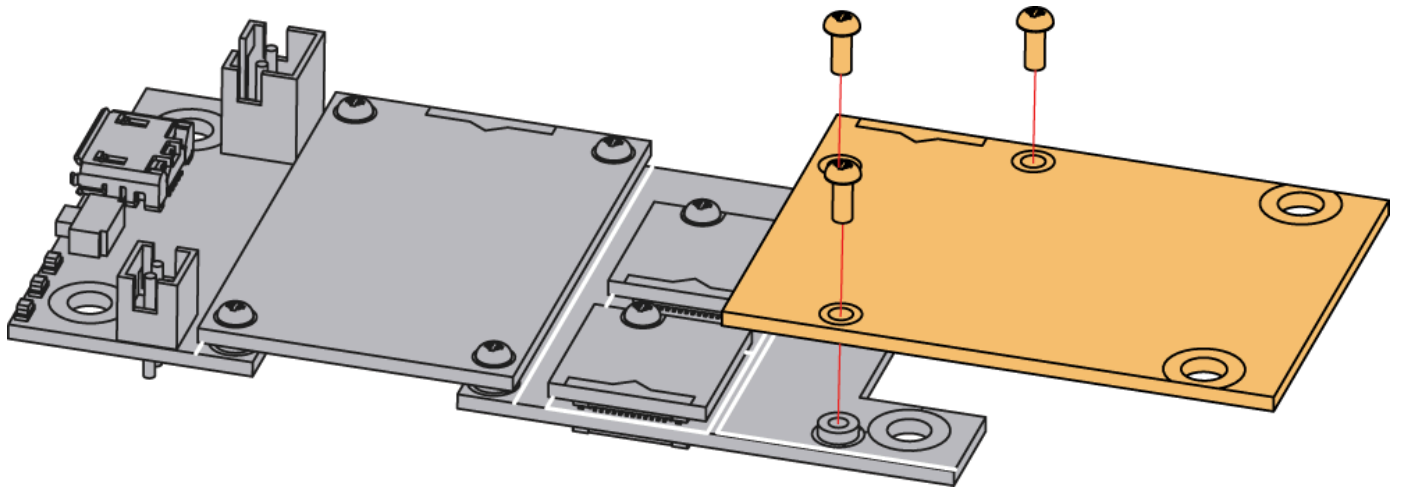
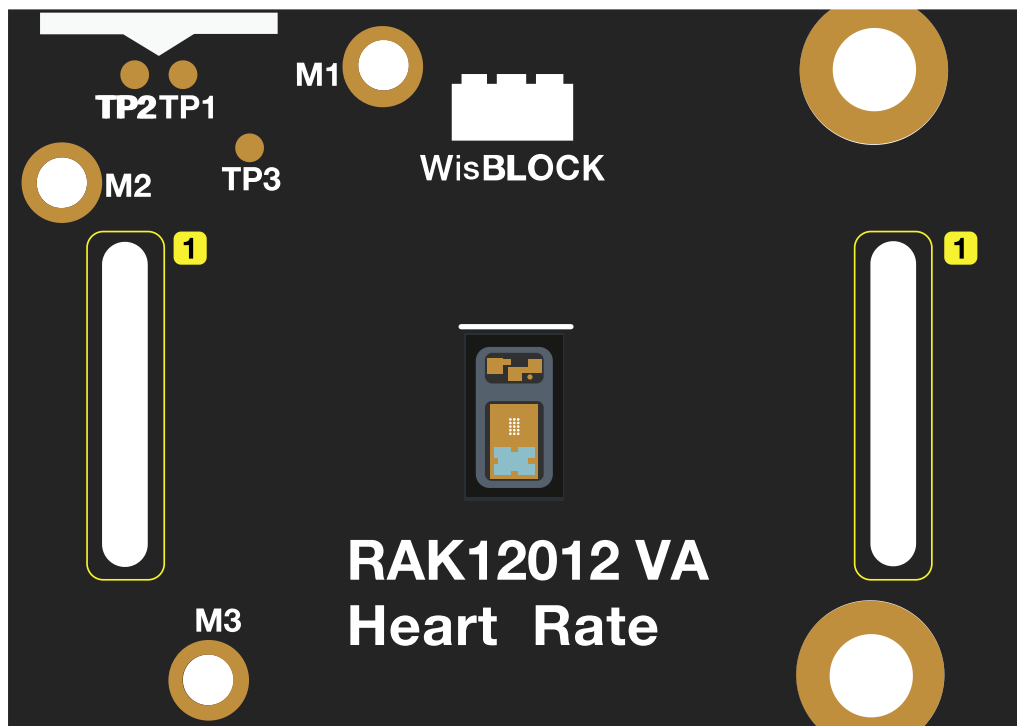


Figure 1: RAK12012 WisBlock Heart Rate Sensor Module Mounting

With the [RAK19008 FPC Cable](#) , RAK12012 can be separated from the WisBlock Base board. The two (2) slots of the board are for the rubber bands. The rubber bands can help to press the sensor against the finger or wrist.



1 Rubber Band Slots

Figure 2: RAK12012 WisBlock Heart Rate Sensor Module Mounting With Rubber Band

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts and its corresponding functions and diagrams. It also covers the electrical and mechanical parameters that include the tabular data of the functionalities and standard values of the RAK12012 WisBlock Heart Rate Sensor Module.

Chipset

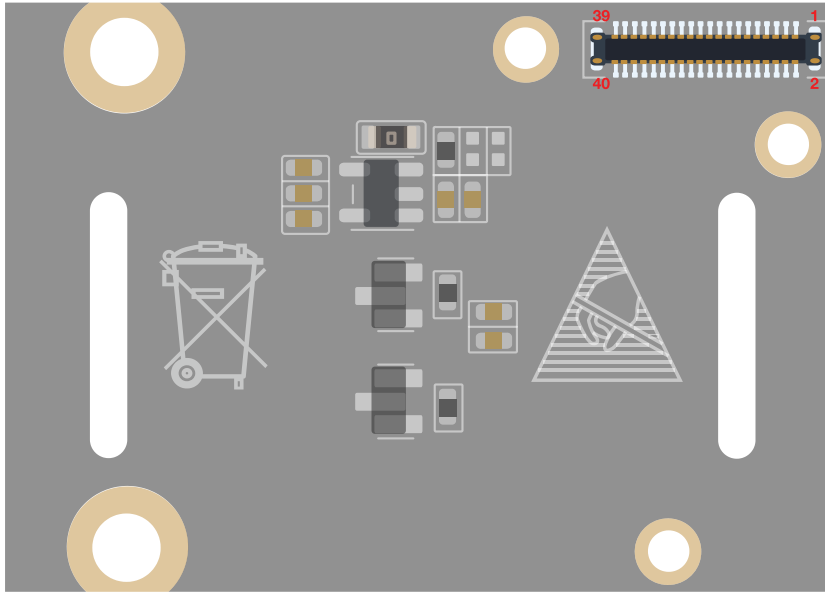
Vendor	Part number
Analog Devices	MAX30102

Pin Definition

The RAK12012 WisBlock Heart Rate Sensor comprises a standard WisBlock connector. The WisBlock connector allows the RAK12012 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition is shown in **Figure 3**.

NOTE

- **I2C** related pins: **INT**, **3V3_S**, and **GND** are connected to WisIO connector.
- **3V3_S** voltage output from the WisBlock Base that powers the RAK12012 module can be controlled by the WisBlock Core via WB_IO2 (WisBlock IO2 pin). This makes the module ideal for low-power IoT projects since the WisBlock Core can totally disconnect the power of the RAK12012 module.



NC	2	1	NC
GND	4	3	GND
3V3_S	6	5	NC
NC	8	7	NC
NC	10	9	NC
NC	12	11	NC
NC	14	13	NC
NC	16	15	NC
NC	18	17	NC
I2C SCL	20	19	I2C SDA
NC	22	21	NC
NC	24	23	NC
NC	26	25	NC
NC	28	27	NC
NC	30	29	NC
INT	32	31	NC
NC	34	33	NC
NC	36	35	NC
NC	38	37	NC
GND	40	39	GND

Figure 3: RAK12012 WisBlock Heart Rate Sensor Module Pinout

Electrical Characteristics

Absolute Maximum Ratings

Parameter	Minimum	Maximum	Unit
3V3_S	-0.3	6	V
1V8	-0.3	2.2	V
ESD (HBM)	-	2.5	KV
Operating Temperature Range	-40	85	°C
Continuous Power Dissipation	-	440	mW

Power Supply Ratings

Symbol	Description	Condition	Min.	Nom.	Max.	Unit
3V3_S	LED Supply Voltage	Input voltage must within this range	3.1	3.3	5	V
1V8	Analog Supply Voltage	Input voltage must within this range	1.7	1.8	2	V
IDD1	Supply Current	SpO2 and HR mode, PW = 215 μ s, 50 sps	-	600	1200	μ A
IDD2	Supply Curren	IR only mode, PW = 215 μ S, 50 sps	-	600	1200	μ A
ISHDN	Supply Current in Shutdown	TA = +25 °C, MODE = 0x80	-	0.7	10	μ A

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanical drawing of the RAK12012 module.

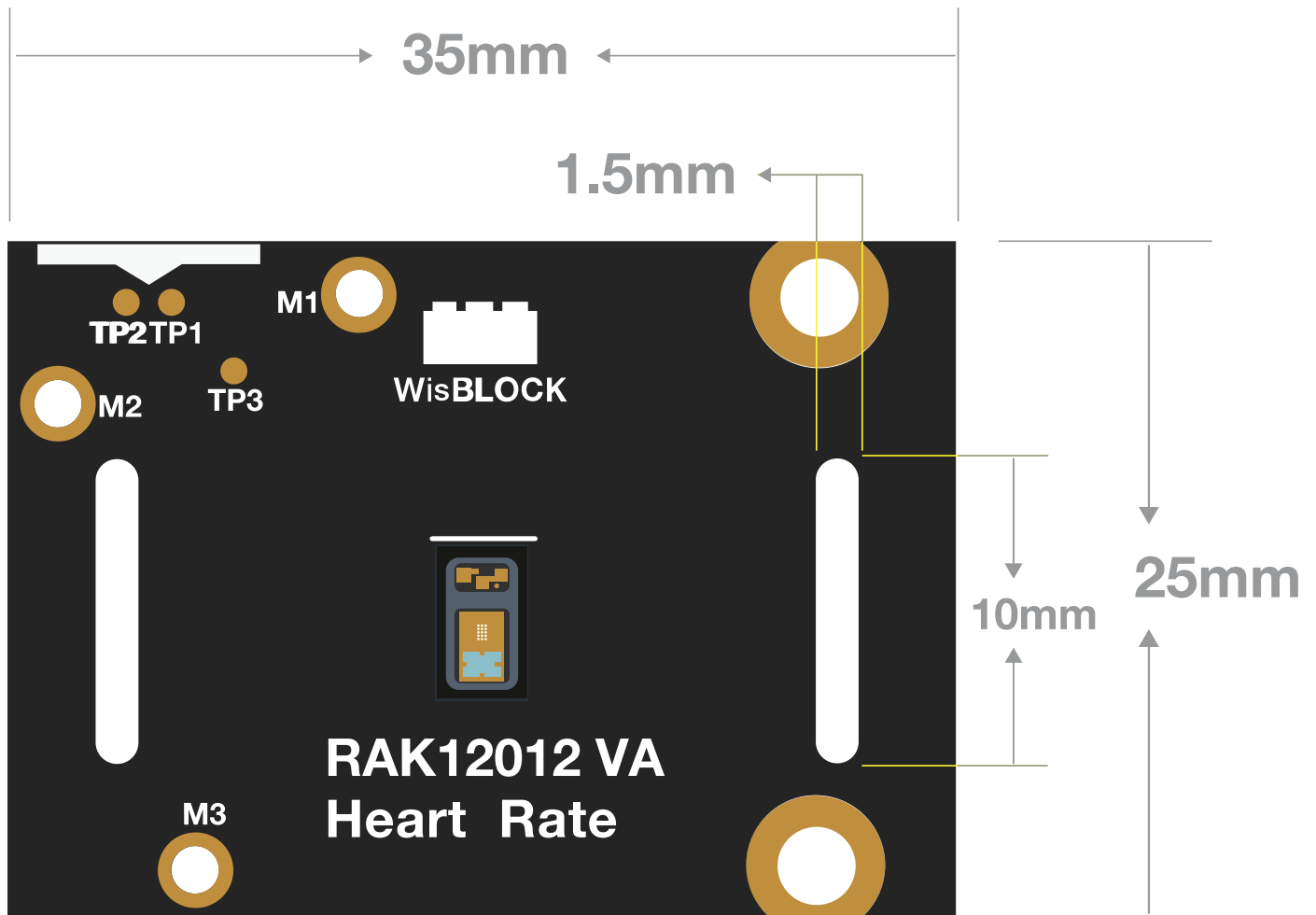


Figure 4: RAK12012 WisBlock Heart Rate Sensor Module Dimensions

WisConnector PCB Layout

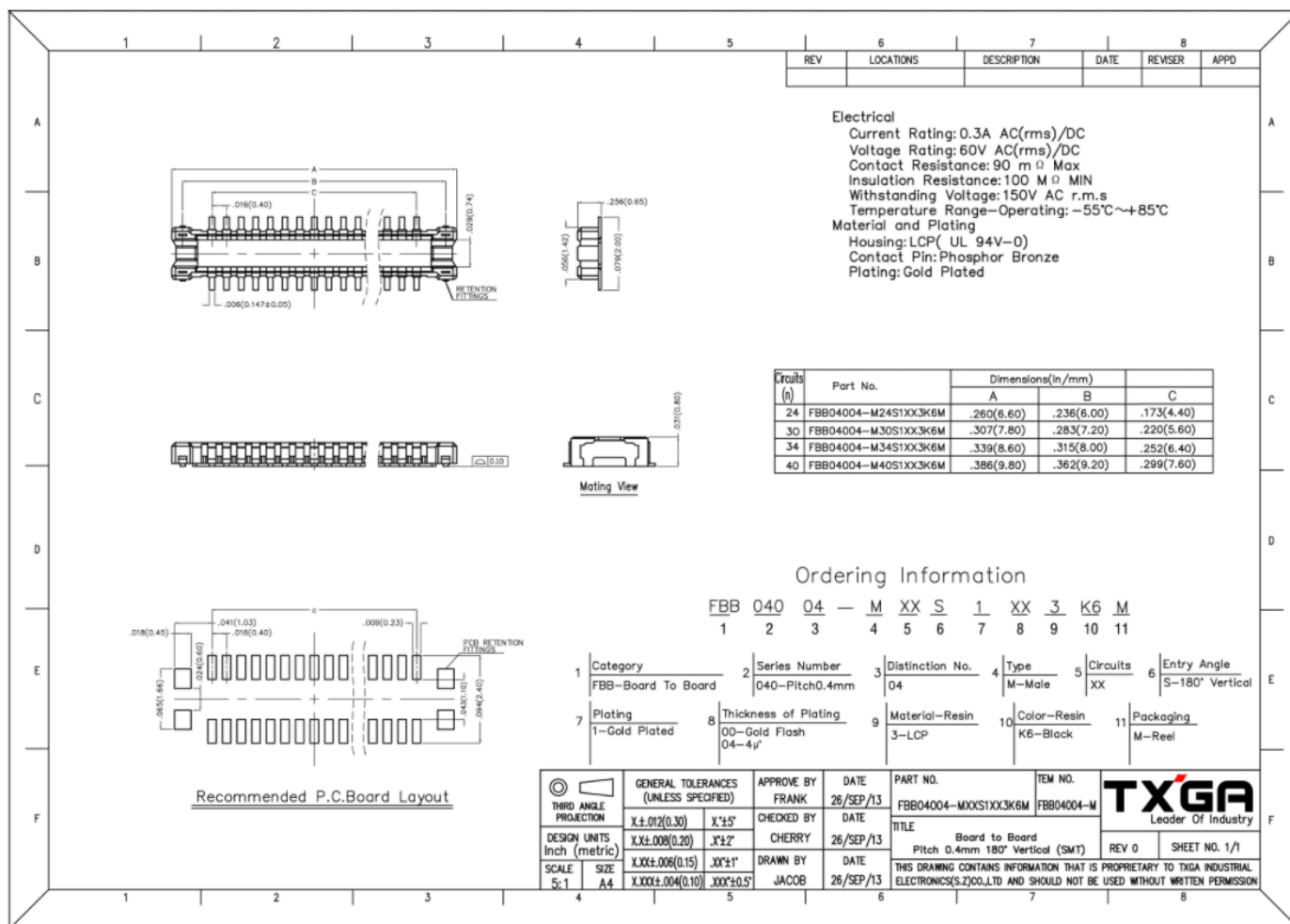


Figure 5: WisConnector PCB Footprint and Recommendations

Schematic Diagram

Power Supply

Figure 6 shows the MAX30102 power supply. The IC uses 1.8 V as voltage supply source generated by SGM2019-1.8YN5G/TR (LDO). The VLED+ has a different dedicated power supply coming from the 3V3_S of the WisBlock Base board.

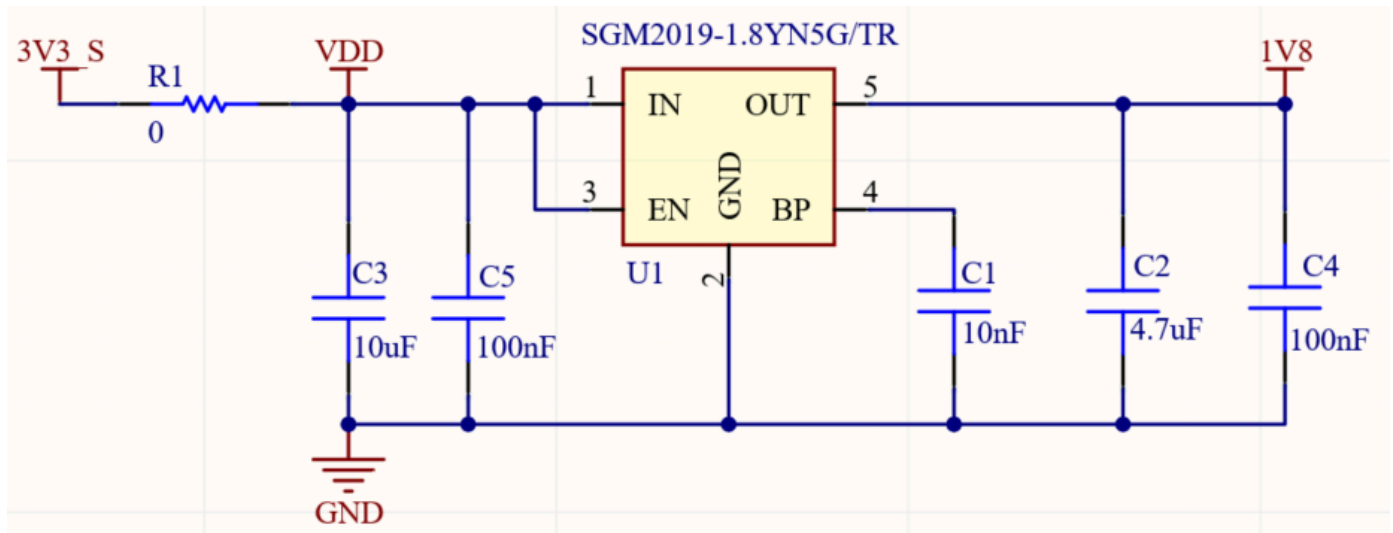


Figure 6: RAK12012 WisBlock Heart Rate Sensor Module Power Supply

Voltage Level Transfer

The MAX30102 IO voltage is 1.8 V and the CPU module voltage is 3.3 V, so the voltage level transfer is needed. Pull-up Resistor for I2C_SCL and I2C_SDA already exist on WisBlock Base board, so there is no need to connect (NC) on IO module such as RAK12012.

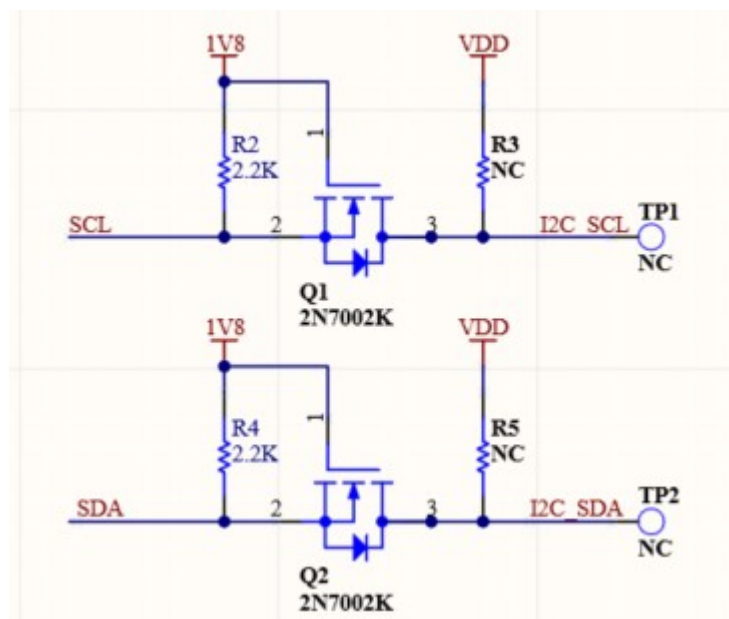


Figure 7: The Voltage Level Transfer Circuit

MAX30102EFD+T Circuit

INT is an active-low interrupt (Open-Drain). This needs to be connected to an external voltage with a pull-up resistor R7.

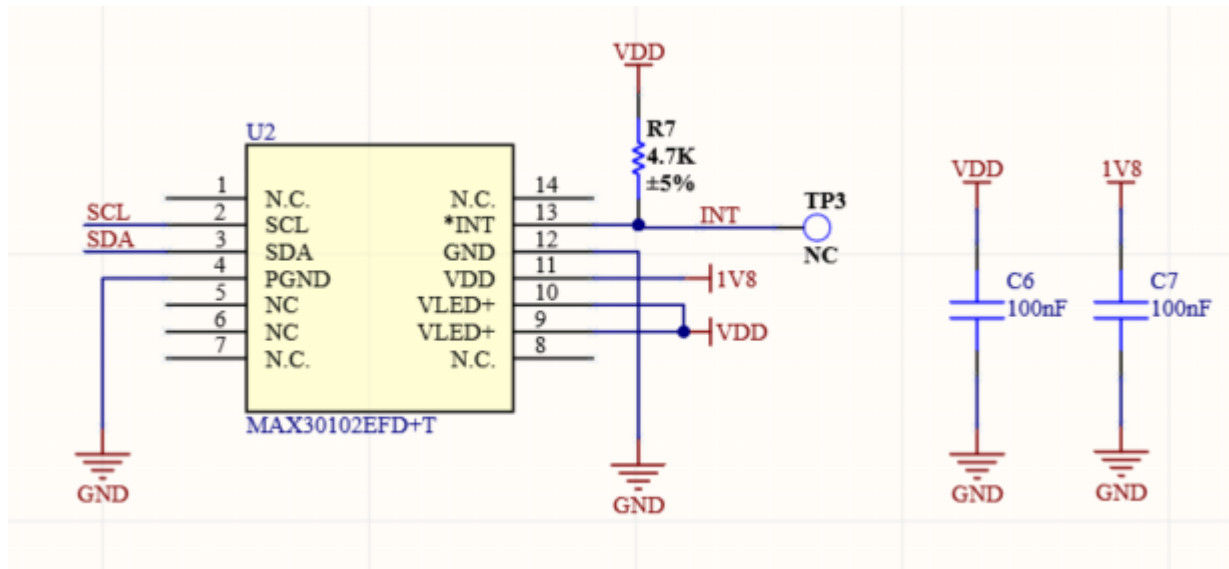


Figure 8: MAX30102EFD+T Circuit

WisConnector

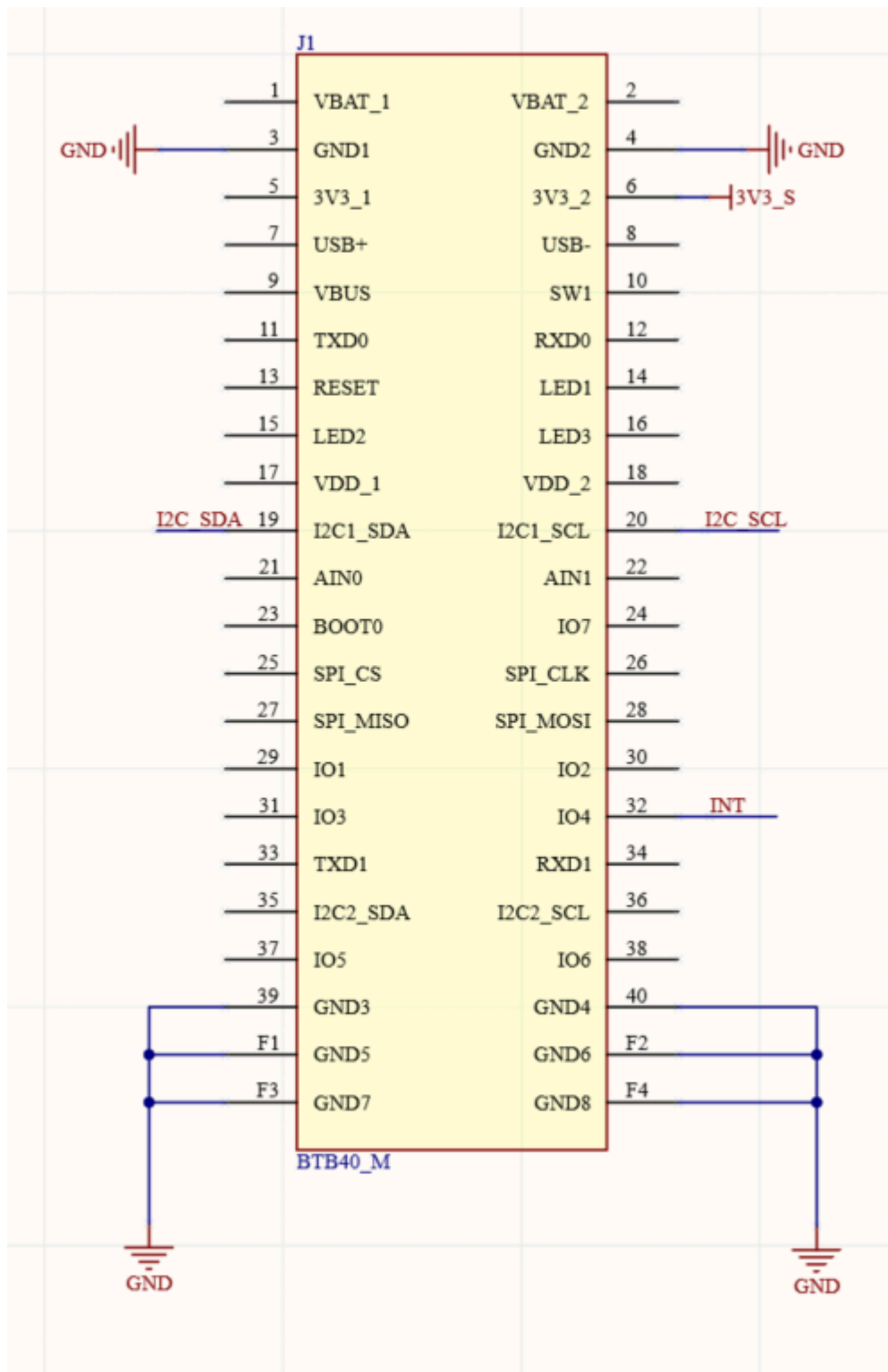


Figure 9: RAK12012 Module WisConnector

NOTE

- The I2C related pins: ****INT ****, **3V3_S**, and **GND** are connected to WisIO connector.

- The **3V3_S** supply can be disconnected to save power via **WB_IO2**.